Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Introduction

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Introduction
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Introduction

1 Introduction

Study focus: Introduction. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000014 · EV0000015

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we'll understand partial differential equations.

Key details

- Introduction to partial differential equations.
- An equation involving independent variable, dependent variable and derivatives of the dependent variable with respect to independent variable is called a differential equation.
- In this case x is the independent variable, y is dependent variable and the derivative of y with respect to x is nothing but dy by dx.
- Differential equations are classified into two based on the number of independent variables present in it.
- The first one ordinary differential equations and partial differential equations.
- In the case of ordinary differential equations, we see that the derivatives are with respect to only one independent variable.
- Also, the unknown function depends on a single variable.
- And X is independent variable.
- In a similar way, in partial differential equations we see that the partial derivatives are with respect to two or more independent variables.
- The unknown function depends on multiple variables.
- In this case u is dependent variable and x and t are independent variables.
- A partial differential equation is an equation that involves partial derivatives of a function with several independent variables.
- As in the case of ordinary differential equations, we have only one independent variable.
- In the case of partial differential equations, we have two or more independent variables.

- Some of the examples of partial differential equations are one dimensional heat conduction equation, Laplask equation, and one dimensional wave equation.
- As we have seen earlier, one of the applications of ordinary differential equations is modeling.
- In a similar manner,.
- Partial differential equations help us in understanding many real world phenomena like waves, those in vibrating strings, heat flow, fluid movement, electromagnetic fields, etc.
- As in the case of ordinary differential equations, the order of a partial differential equation is the order of the.

Example or application from the lecture

- Before we understand what is a partial differential equation, let's look at what is a differential equation.
- As an example, dy by dx plus y equals e power x.
- Let's understand the classification of differential equations.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 05:00–09:59

Highest order partial derivative present in it.

Key details

- The partial derivatives.
- As you can see, some of the examples to answer the order and degree of a partial differential equation.
- Then pqr s t are denoted by dou z by doux, dou z by dou y dou square z divided by dou x square, dou square z divided by dou x dou y and dou square z divided by dou y square.
- So based on the standard notations, an equation containing X, Y, Z, P, and Q defines a first order partial differential equation.
- Remember, this first order partial differential equation is said to be linear if it is linear in P and Q.
- It means we cannot have the product of P and Q, we cannot have P squared, we cannot have Q square, and so on.
- An equing XYZ PQRST defines a second order partial differential equation.
- That is G of XYZ PQRST equals zero.
- If it is linear in PQRST.

Example or application from the lecture

- In the first example, you can see order is also one, degree is also one.
- In the second example, order and degree are two, one, and one respectively.
- In the third example, order and degree are two and one respectively.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Introduction transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Introduction__9ffc6e5f7e376549.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.

2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00006	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...
SRC00007	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...
SRC00009	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0029. Sidecar variants compare...

C Revision and retrieval practice